PLATFORM CONCEPT OVERVIEW

Florida New Construction

Sales Velocity, Pricing & Market Analytics Platform

1. Executive Summary

This document outlines the product concept for an interactive analytics platform focused on tracking newly constructed residential subdivisions across 23 target counties in Florida. The platform provides real-time visibility into monthly sales velocity (absolute unit count and absorption rate), median sales price, lot size mix, builder market share, and resale comparable inventory, all segmented by product type (Detached Single-Family Home and Townhome).

The 23 counties in scope are: Saint Lucie, Indian River, Brevard, Osceola, Orange, Seminole, Lake, Sumter, Polk, Hillsborough, Manatee, Pasco, Hernando, Citrus, Marion, Volusia, Flagler, St. Johns, Clay, Baker, Alachua, Duval, and Nassau. These counties represent the core Central, Northeast, and Gulf Coast Florida markets with the highest concentration of new construction activity.

The platform is designed as a layered, drill-down experience: users begin with a regional heat map of the target counties, narrow into county-level views featuring builder presence and resale comparables, explore individual subdivisions with lot size mix and velocity breakdowns, and examine historical performance data from subdivision inception to the present month. Additional platform-wide features include a builder leaderboard, months of supply tracking, incentive/concession monitoring, price-per-square-foot analysis, and automated alerts.

2. Feature Summary

The following table provides a comprehensive breakdown of all platform features organized by category. This serves as the feature inventory for sprint planning and prioritization.

2.1 Core Map & Navigation

Feature	Description	Platform Layer
Regional Heat Map	Choropleth map of 23 FL counties colored by selected metric. Landing view for the platform.	Layer 1
Metric Toggle	Switch heat map between Velocity (absorption rate) and Median Sales Price.	Layer 1
Product Type Toggle	Switch between Detached SFH and Townhome segments. Independent of metric toggle (4 combo views).	Layer 1
County Drill-Down	Click any county polygon to zoom into county view with subdivision pins.	Layer 1 → Layer 2
Subdivision Pins	Map markers for each qualifying subdivision, color-coded by velocity tier (high/med/low).	Layer 2
Subdivision Drill-Down	Click any pin or sidebar row to open subdivision snapshot.	Layer 2 → Layer 3
Historical View	Time-series charts from subdivision inception to current month.	Layer 3 → Layer 4

Feature	Description	Platform Layer
Breadcrumb Navigation	Persistent breadcrumb trail (FL Map › County › Subdivision › History) for back-navigation.	All Layers

2.2 Velocity & Pricing Metrics

Feature	Description	Platform Layer
Absolute Sales Count	Total closed transactions (warranty deeds recorded) per subdivision per calendar month.	Layers 2–4
Absorption Rate	Monthly closed sales as a percentage of available inventory at month start.	Layers 1–4
Median Sales Price	Median closed transaction price per month, segmented by Detached SFH and Townhome.	Layers 1–4
Price Per Square Foot	Median $/sqft computed from closed transactions. Enables apples-to-apples comparison across subdivisions with different home sizes.	Layers 3–4
Month-over-Month Deltas	Directional change indicators (▲/▼ with green/red color) for velocity and price metrics.	Layers 1–4
Months of Supply	Active inventory divided by monthly velocity. Displayed at county and subdivision level. Industry-standard gauge for over/undersupply.	Layers 2–3

2.3 Lot Size & Unit Mix

Feature	Description	Platform Layer
Lot Size Breakdown	Display lot dimensions (e.g., 50’ x 120’, 40’ x 120’) per subdivision. Sourced from permits, plat records, parcel GIS, and MLS.	Layer 3
Velocity by Lot Size	Monthly sales count broken down by lot size tier. E.g., 6 total sales = 4 on 50x120 lots + 2 on 40x120 lots.	Layers 3–4
Median Price by Lot Size	Median closed price segmented by lot size tier within a subdivision.	Layers 3–4
Lot Size Inventory	Remaining available lots broken down by lot size tier.	Layer 3

2.4 Builder Presence & Market Share

Feature	Description	Platform Layer
Submarket Builder Presence	Snapshot of each builder’s active pipeline in the county. Shows units controlled, percentage of total county inventory, and number of active subdivisions.	Layer 2

Feature	Description	Platform Layer
Builder Market Share	Calculated as: (Builder's Available Inventory in County / Total Available Inventory in County) × 100.	Layer 2
Builder Leaderboard	Top-level view ranking builders by total market share, velocity, and median price across all 23 counties. Allows builder-first (vs. geography-first) analysis.	New Top-Level View
Builder Filter	Filter subdivisions within a county by builder name.	Layer 2

2.5 Resale Comparable Inventory

Feature	Description	Platform Layer
County Resale Inventory Count	Total active resale listings in the county for homes built within the last 5 years. Sourced from MLS and/or Zillow. Displayed alongside new construction metrics.	Layer 2
Resale Absorption Rate	Monthly resale closings for comparable homes (built within 5 years) as a percentage of active resale listings. Enables direct velocity comparison: new construction vs. resale.	Layer 2
Resale Median Price	Median closed price for comparable resale homes (built within 5 years), segmented by Detached SFH and Townhome.	Layer 2
Subdivision-Level Resale Context	For a selected subdivision, display resale inventory and velocity for comparable homes (built within 5 years) within the same county. Provides competitive context.	Layer 3
New vs. Resale Comparison Panel	Side-by-side display of new construction and resale metrics: inventory count, velocity, median price, and months of supply. Visual indicator showing which segment is outperforming.	Layers 2–3

2.6 Incentive & Concession Tracking

Feature	Description	Platform Layer
Concession Rate	Percentage of monthly closings that included seller concessions (sourced from MLS concession fields).	Layers 3–4
Median Concession Amount	Median dollar value of concessions on applicable closings.	Layers 3–4
Concession Trend	Month-over-month trend line for concession rate. Serves as a leading indicator of builder pricing pressure before median price drops.	Layer 4

2.7 Alerts & Watchlists

Feature	Description	Platform Layer
Watchlist	Users can bookmark specific counties or subdivisions to a personal watchlist for quick access.	All Layers
Velocity Alert	Email/push notification when a watched subdivision's absorption rate crosses a user-defined threshold (e.g., drops below 5%).	Background Service
Price Alert	Notification when median price shifts more than ±X% month-over-month for a watched entity.	Background Service
Inventory Alert	Notification when months of supply exceeds a user-defined ceiling (e.g., >12 months), signaling potential oversupply.	Background Service

2.8 Data Export

Feature	Description	Platform Layer
CSV Export	Download raw historical data for any subdivision or county as a CSV file.	Layer 4
PDF Report	Formatted summary report with charts for a selected subdivision or county.	Layer 4

3. Product Vision & Objectives

3.1 Vision Statement

Deliver the most intuitive and data-rich interactive map for tracking new construction sales activity in Florida, enabling MAS Development to make faster, more informed decisions at the subdivision level.

3.2 Core Objectives

1. Provide a regional heat map across 23 target counties with county-level granularity for sales velocity and median sales price.
2. Enable seamless drill-down from state to county to subdivision to historical trend views.
3. Segment all pricing and velocity data by product type (Detached SFH and Townhome) and by lot size where data permits.
4. Track builder market share and active pipeline at the county (submarket) level.
5. Compare new construction velocity and pricing against resale comparable inventory (homes built within 5 years) at both the county and subdivision level.
6. Surface leading indicators of pricing pressure through incentive/concession tracking.

7. Track historical performance since subdivision inception to reveal lifecycle trends.
8. Enable proactive monitoring through alerts and watchlists.
9. Integrate public, MLS, and third-party data sources with monthly refresh cadence.

4. Key Definitions

4.1 Newly Developed Subdivision

A precise, repeatable definition is critical for data consistency. The platform will classify a subdivision as "newly developed" based on the following composite criteria:

Criterion	Definition
Plat Recording Date	The subdivision plat was recorded with the county clerk within the last 10 years from the current reporting month.
Active Building Permits	At least one new residential building permit has been issued within the subdivision boundaries in the trailing 24 months.
Builder Activity	A licensed homebuilder is listed as the seller/grantor on at least one closed transaction within the trailing 12 months.
Product Type	The subdivision contains at least one lot zoned and permitted for single-family detached or townhome residential construction.
Completion Threshold	Fewer than 95% of total platted lots have received a Certificate of Occupancy (CO), indicating the community has not fully built out.

Exclusion Criteria: Subdivisions that are 95% or more built out (by CO count vs. platted lots), have had no building permit activity for 24+ months, or are zoned exclusively for commercial or multi-family (5+ units) use will be excluded from the dataset.

4.2 Sales Velocity

Sales velocity is reported using two complementary metrics:

Metric	Definition
Absolute Sales Count	The total number of closed residential transactions (warranty deeds recorded) within a given subdivision during a single calendar month.
Absorption Rate	The percentage of remaining available inventory (unsold lots or completed-but-unsold homes) that sold during the calendar month. Calculated as: (Monthly Closed Sales / Total Available Inventory at Month Start) x 100.

4.3 Median Sales Price

The median (middle value) of all closed transaction prices within a given subdivision, county, or regional scope during a calendar month. The median is preferred over the arithmetic mean

because it is less sensitive to outlier transactions (e.g., model homes, spec home discounts, or luxury upgrades) and provides a more representative measure of typical pricing. Prices are segmented by product type:

- **Detached SFH:** Single-family detached homes on individual lots.
- **Townhome:** Attached residential units sharing one or more walls, typically with individual lot ownership.

4.4 Builder Market Share

A builder's market share within a county is calculated as the number of available inventory units controlled by that builder divided by the total available inventory in the county, expressed as a percentage. "Available inventory" includes all platted lots that have not yet received a Certificate of Occupancy, plus completed homes that have a CO but have not yet closed with a buyer (builder-owned standing inventory).

Formula: (Builder's Available Inventory in County / Total County Available Inventory) × 100

4.5 Resale Comparable Inventory

Resale comparable inventory is defined as active resale listings (not new construction) for residential properties that were built within the last 5 years, located within the same county as the new construction subdivisions being analyzed. This filter ensures the resale comparison pool reflects homes of similar age, construction quality, and buyer appeal. Listings are sourced from MLS and/or Zillow data feeds.

- **Geographic Scope:** County-wide (all active resale listings within the county boundary).
- **Age Filter:** Year built must be within the last 7 years from the current reporting month.
- **Product Segmentation:** Resale listings are segmented by Detached SFH and Townhome to enable direct comparison with corresponding new construction segments.

4.6 Lot Size Classification

Lot size refers to the platted lot dimensions (frontage width × depth) for a given parcel within a subdivision. Lot sizes are expressed in the industry-standard format (e.g., 50' x 120'). The platform will group lots into tiers defined by frontage width (e.g., 30', 35', 40', 50', 60', 70', 80'+) and attribute velocity and pricing metrics to each tier.

Lot size data will be sourced through a multi-layered approach, prioritized as follows:

10. **MLS Lot Dimensions Field:** Many MLS listings include lot size or lot dimensions in structured fields. This is the most accessible source and should be checked first.
11. **Recorded Plat Maps:** The original subdivision plat filed with the county clerk defines lot dimensions for every lot. Plat maps can be parsed programmatically if digitized, or manually cataloged per subdivision.
12. **County Parcel/GIS Data:** County property appraiser GIS layers typically include parcel polygons from which lot frontage and depth can be computed geometrically.
13. **Building Permit Records:** Some municipalities include lot size or lot number references in permit filings, which can be cross-referenced with plat maps.

14. **Manual Builder Spec Sheets:** As a fallback, publicly available builder site plans and community maps can be used to catalog lot size tiers per subdivision.

4.7 Months of Supply

Months of supply is the ratio of active inventory to the current pace of sales, calculated as: Total Available Inventory / Average Monthly Velocity (trailing 3-month average). This metric is computed at both the county level and subdivision level. A months-of-supply figure below 6 generally indicates an undersupplied market; above 9 indicates potential oversupply.

4.8 Concession Rate

The concession rate is the percentage of closed transactions in a given month where the seller provided a financial concession (e.g., closing cost credits, rate buydowns, or price reductions at closing). This is sourced from MLS concession fields. The concession rate serves as a leading indicator of builder pricing pressure: a rising concession rate often precedes a decline in median sales price.

5. Layered Product Architecture

The platform is designed as a four-layer drill-down experience with a supplementary builder leaderboard view. Each layer adds specificity while maintaining contextual navigation back to higher levels.

5.1 Layer 1 — Regional Heat Map (Landing View)

Purpose: Provide an at-a-glance view of new construction sales activity across the 23 target Florida counties.

- **Default View:** Choropleth heat map colored by sales velocity (absorption rate) for Detached SFH.
- **Toggle Controls:** Users can switch between (a) Velocity vs. Median Sales Price, and (b) Detached SFH vs. Townhome. This creates four possible heat map views.
- **Color Scale:** Gradient from cool (low activity/price) to hot (high activity/price) with a dynamic legend.
- **Hover Tooltip:** County name, current month metric value, month-over-month change.
- **Interaction:** Click on any county polygon to drill into Layer 2.

5.2 Layer 2 — County View (Subdivision Map + Builder Presence + Resale Comps)

Purpose: Display all newly developed subdivisions within the selected county, builder market share breakdown, and resale comparable inventory context.

5.2.1 Subdivision Map Panel

- **Map Pins/Polygons:** Each pin represents a qualifying subdivision. Pins are color-coded by velocity tier (high/medium/low).
- **Sort/Filter Panel:** Sort subdivisions by velocity, price, or builder. Filter by product type, price range, lot size, or builder name.
- **Interaction:** Click on a subdivision pin to drill into Layer 3.

5.2.2 County Summary Sidebar

The sidebar displays county-level aggregate metrics:

- **Total Subdivisions Tracked:** Count of qualifying newly developed subdivisions in the county.
- **County-Wide Velocity:** Aggregate absorption rate and absolute sales count.
- **County-Wide Median Price:** Median closed price by product type (Detached SFH and Townhome).
- **Months of Supply:** County-level months of supply for new construction inventory.

5.2.3 Submarket Builder Presence Panel

A dedicated panel (collapsible section or tab within the county sidebar) showing each builder's active pipeline in the county:

Data Point	Description
Builder Name	Name of each builder with active inventory in the county.
Available Inventory (Units)	Number of unsold lots + builder-owned standing inventory controlled by this builder.
Market Share (%)	Builder's inventory as a percentage of total county available inventory.
Active Subdivisions	Number of subdivisions in the county where this builder has active inventory.
Avg. Monthly Velocity	Builder's trailing 3-month average sales count within the county.

Example: If Orange County has 1,000 total available new construction units and DR Horton controls 400 of them, DR Horton's market share is displayed as 40%. The panel would also show that DR Horton has inventory in 8 subdivisions and is averaging 15 sales per month across the county.

5.2.4 Resale Comparable Inventory Panel

A "New vs. Resale" comparison panel displayed within the county view sidebar, showing side-by-side metrics:

Metric	New Construction	Resale (Built <5 Yrs)
Active Inventory	Total unsold new construction units	Active resale listings (year built within 7 years)
Monthly Velocity	Absorption rate for new construction	Absorption rate for comparable resale
Median Price — Detached SFH	New construction median	Resale comparable median

Metric	New Construction	Resale (Built <5 Yrs)
Median Price — Townhome	New construction median	Resale comparable median
Months of Supply	New construction MoS	Resale comparable MoS

5.3 Layer 3 — Subdivision Snapshot (Current Month)

Purpose: Provide the most recent month's key performance metrics for the selected subdivision, including lot size mix and resale context.

Data Point	Description
Subdivision Name	Official plat name as recorded with the county.
Builder(s)	Primary builder(s) active in the subdivision.
Total Platted Lots	Total number of lots in the recorded plat.
Lots Remaining	Lots without a Certificate of Occupancy.
Current Month Velocity (Absolute)	Number of closed sales this month.
Current Month Velocity (Absorption)	Absorption rate for this month.
Median Sales Price — Detached SFH	Median closed price for detached homes this month.
Median Sales Price — Townhome	Median closed price for townhomes this month.
Price Per Square Foot	Median $/sqft for closed transactions this month.
Months of Supply	Remaining inventory / trailing 3-month velocity.
Concession Rate	% of this month's closings with seller concessions.
Median Concession Amount	Median dollar value of concessions this month.

5.3.1 Lot Size Mix & Velocity by Lot Size

If lot size data is available for the subdivision, an additional panel displays:

Data Point	Description
Lot Size Tiers Available	List of lot dimensions in the subdivision (e.g., 40'x120', 50'x120', 60'x120').
Inventory by Lot Size	Remaining available lots per tier.
Monthly Sales by Lot Size	Closed sales this month per tier (e.g., 4 sales on 50x120, 2 sales on 40x120).
Median Price by Lot Size	Median closed price per lot size tier this month.
$/Sqft by Lot Size	Median price per square foot per lot size tier.

If lot size data is not available for a specific subdivision, this panel displays a notice: “Lot size data not available for this subdivision” with a link to request manual classification. User can also manually add this information.

5.3.2 Resale Context at Subdivision Level

A compact comparison widget showing resale inventory metrics for comparable homes (built within 7 years) county-wide, positioned alongside the new construction metrics to provide competitive context. This uses the same New vs. Resale format as the county-level panel but scoped to the context of evaluating a single subdivision’s positioning against the broader resale market.

Interaction: A “View History” button navigates to Layer 4.

5.4 Layer 4 — Subdivision Historical Performance

Purpose: Visualize the full lifecycle of the subdivision from inception to the current month.

- **Chart 1 — Monthly Sales Velocity:** Dual-axis chart: bars = absolute sales count (left axis), line = absorption rate % (right axis).
- **Chart 2 — Monthly Median Sales Price:** Separate lines for Detached SFH and Townhome.
- **Chart 3 — Velocity by Lot Size (if data available):** Stacked bar chart showing monthly sales broken down by lot size tier.
- **Chart 4 — Concession Trend:** Line chart showing concession rate % and median concession amount over time. Leading indicator overlay.
- **Data Range:** From the month of the first recorded closed sale in the subdivision through the most recent reporting month.
- **Summary Statistics:** Cumulative units sold, average monthly velocity, median sales price, price trend direction, estimated months of remaining inventory, and lifetime concession rate.
- **Export Options:** Download historical data as CSV or formatted PDF report.

5.5 Builder Leaderboard (Top-Level View)

Purpose: Provide a builder-first view ranking builders by performance across all 23 target counties. Accessible from the global navigation bar alongside the heat map.

Data Point	Description
Builder Name	Name of the builder.
Total Market Share (%)	Builder’s available inventory as a % of total inventory across all 23 counties.
Total Available Inventory	Sum of unsold units across all counties.
Active Counties	Number of counties where the builder has active inventory.
Active Subdivisions	Total subdivision count across all counties.
Trailing 3-Mo Velocity	Average monthly closed sales across all builder’s subdivisions.

Data Point	Description
Median Price — SFH	Median closed price for detached homes across builder's portfolio.
Median Price — TH	Median closed price for townhomes across builder's portfolio.
Concession Rate	Portfolio-wide concession rate.

The leaderboard is sortable by any column and filterable by county. Clicking a builder name drills into a builder profile page showing county-by-county and subdivision-by-subdivision breakdown.

6. User Experience Flow

The following summarizes the primary navigation paths:

6.1 Geography-First Path (Primary)

Step	View	User Action	Result
1	Regional Heat Map	Selects metric toggle (Velocity or Median Price) and product type (Detached or Townhome)	Heat map recolors to reflect selected metric/segment
2	Regional Heat Map	Clicks on a county	Zooms into county; transitions to Layer 2
3	County View	Reviews builder presence panel and resale comparison	Understands builder landscape and resale competitive context
4	County View	Clicks on a subdivision pin or sidebar row	Opens Layer 3 snapshot panel
5	Subdivision Snapshot	Reviews velocity, pricing, lot size mix, and resale context	Full current-month picture of subdivision performance
6	Subdivision Snapshot	Clicks "View History"	Navigates to Layer 4 historical charts
7	Historical Performance	Explores time-series charts, reviews concession trends, exports data	Full lifecycle view from inception to current month

6.2 Builder-First Path (Secondary)

Step	View	User Action	Result
1	Builder Leaderboard	Sorts by market share or velocity	Identifies builders with largest presence or fastest sales

Step	View	User Action	Result
2	Builder Leaderboard	Clicks on a builder name	Opens builder profile with county and subdivision breakdown
3	Builder Profile	Clicks on a specific subdivision	Navigates to Layer 3 subdivision snapshot

7. Heat Map Toggle Matrix

The regional heat map supports four distinct views controlled by two independent toggles:

Toggle A: Metric	Toggle B: Product Type	Heat Map Shows	Color Gradient Represents
Velocity	Detached SFH	County-level absorption rate for detached homes	Low absorption (cool) to High absorption (hot)
Velocity	Townhome	County-level absorption rate for townhomes	Low absorption (cool) to High absorption (hot)
Median Sales Price	Detached SFH	County-level median closed price for detached homes	Low price (cool) to High price (hot)
Median Sales Price	Townhome	County-level median closed price for townhomes	Low price (cool) to High price (hot)

8. Data Architecture & Integration

8.1 Data Pipeline Overview

The platform requires a robust ETL pipeline that ingests data from multiple sources, normalizes it, matches records to subdivisions, and refreshes on a monthly cadence. The high-level pipeline stages are: (1) Ingestion from public APIs, MLS feeds, and third-party vendors, (2) Cleaning, normalization, and lot size classification, (3) Geocoding and subdivision matching, (4) Metric computation including builder market share and resale comparables, and (5) Storage in a queryable analytics database.

8.2 Recommended Third-Party Tools & Integration Points

Function	Recommended Tool	Integration Layer	Notes
Interactive Map Rendering	Mapbox GL JS	Frontend	High-performance WebGL map with custom polygon layers, heat map support, and smooth zoom transitions.
Alternative Map Option	Google Maps Platform (Maps JS API)	Frontend	Broader familiarity; strong geocoding. Higher per-load cost at scale.
County Boundary Polygons	US Census Bureau TIGER/Line Shapefiles	Data Pipeline / Frontend	Free GeoJSON county boundaries; pre-processed as vector tiles.
Geocoding & Address Normalization	Google Maps Geocoding API or Mapbox Geocoding	Data Pipeline	Convert recorded addresses to lat/lng for subdivision pin placement.
Charting & Visualization	Recharts or D3.js	Frontend	Time-series charts for Layer 4. Recharts for speed; D3 for customization.
Backend API Framework	Node.js (Express/Fastify) or Python (FastAPI)	Backend	RESTful or GraphQL API serving computed metrics to frontend.
Database (Analytics Store)	PostgreSQL + PostGIS	Backend	PostGIS for spatial queries. Pair with TimescaleDB for time-series.
Cloud Hosting	AWS (S3, Lambda, RDS) or GCP	Infrastructure	S3 for static assets; Lambda for serverless ETL; RDS for managed Postgres.
ETL Orchestration	Apache Airflow or dbt	Data Pipeline	Airflow for scheduling; dbt for transformation and data modeling.
Caching Layer	Redis or AWS ElastiCache	Backend	Cache county/subdivision metrics to reduce database load.
Notification Service	AWS SNS + SES or SendGrid	Backend	Email/push delivery for alert and watchlist notifications.

9. Potential Public Data APIs & Sources

9.1 Property Transaction & Sales Data

Source	Data Provided	Access Method	Coverage
Florida Dept. of Revenue (Property Tax Data)	Recorded sales (deed transfers), assessed values, property characteristics	Annual bulk NAL files (downloadable)	All FL counties; 23 target counties extracted.
County Property Appraiser Websites	Parcel data, sales history, building permits, legal descriptions	Varies by county (some APIs, most CSV/bulk)	County-specific; quality varies.
County Clerk of Court (Official Records)	Recorded warranty deeds, plat maps, document images	Varies: some offer searchable online portals	County-specific; authoritative for deed dates and prices.
ATTOM Data Solutions API	Transactions, deed records, AVM, building permits, foreclosures	RESTful API (commercial, paid)	Nationwide; FL coverage is strong.

9.2 Building Permit Data

Source	Data Provided	Access Method	Coverage
US Census Bureau — Building Permits Survey	Monthly new residential permits by county	API (Census Data API) + bulk CSV	All FL counties; filter to 23 targets. ~2 month lag.
HUD SOCDS Building Permits Database	Annual/monthly permit counts by geography and type	Web query tool + downloadable files	Metro and county level.
Local Municipality Permit Portals	Individual permit records with parcel, contractor, scope, and sometimes lot size details	Varies (web portals, some API access)	Jurisdiction-specific; most granular for permit-to-subdivision matching.

9.3 Geographic & Boundary Data

Source	Data Provided	Access Method	Coverage
US Census Bureau TIGER/Line Shapefiles	County boundaries, census tract/block polygons, road network	Bulk download (Shapefiles/GeoJSON)	Nationwide; updated annually.
Florida Geographic Data Library (FGDL)	Parcels, land use, zoning, plat boundaries	Bulk download from UF GeoPlan Center	Statewide.

Source	Data Provided	Access Method	Coverage
County GIS / Open Data Portals	Parcel polygons, subdivision plat boundaries, zoning overlays, lot dimensions	ArcGIS REST APIs, GeoJSON downloads	County-specific; many FL counties publish via ArcGIS Online. Critical source for lot size data.

9.4 MLS Transaction Data

Source	Data Provided	Access Method	Coverage
Stellar MLS (RESO Web API)	Closed transactions, listing data, property type, subdivision, builder, lot size, concessions, days on market	RESO Web API (licensing agreement)	Central/SW FL: Orange, Osceola, Seminole, Lake, Polk, Hillsborough, Manatee, Pasco, Hernando, Brevard, Volusia, and others.
NE Florida MLS / My Florida Regional MLS	Closed transactions, listing data, property details	RESO Web API or RETS feed (licensing agreement)	NE FL: Duval, St. Johns, Clay, Nassau, Baker, Alachua, Flagler.
RESO Web API (Standard)	Standardized data schema across MLS boards	RESTful API per RESO Data Dictionary	Industry standard; consistent field naming.
Bridge Interactive API (MLS Grid)	Aggregated MLS data across multiple boards	RESTful API (commercial agreement)	Multi-board aggregator; simplifies integration.

MLS Integration Note: MLS data serves as the primary source for (a) closed transaction prices and dates, (b) property type classification, (c) lot size/dimensions where available in listing fields, (d) seller concession amounts, and (e) resale comparable inventory (active listings for homes built within 5 years). County clerk deed recordings serve as a secondary validation layer. MLS access requires a broker sponsor or data licensing agreement.

MLS integration will bring us a competitive edge when analyzing the market. When looking into how to integrate the data, please let us know what you need on our end to make this happen. Priority for us.

9.5 Resale Inventory & Enrichment Sources

Source	Data Provided	Access Method	Coverage
Zillow API / Zillow Research Data	Active listings, Zestimates, market trends, year built filter	API (partnership agreement) or public CSV datasets	Nationwide; useful for resale inventory counts and pricing.
Redfin / Realtor.com (where available)	Active listings, recently sold, market stats	API (partnership) or public data downloads	Supplemental resale data.

Source	Data Provided	Access Method	Coverage
Florida DBPR	Licensed contractor and builder registry	Web search / downloadable files	Statewide; builder identification.
FHFA House Price Index API	Regional house price indices	RESTful API (free)	Metro/state level; contextual benchmarking.

9.6 Lot Size Data Sources (Multi-Layered Approach)

Lot size data is critical to the platform and will be sourced through a prioritized multi-layered approach:

Priority	Source	Data Available	Reliability	Notes
1 (Primary)	MLS Lot Dimensions Field	Lot frontage, depth, and/or acreage from listing data	High (when populated)	Not all listings include lot dimensions. Coverage varies by MLS board and listing agent.
2	Recorded Plat Maps (County Clerk)	Exact lot dimensions for every lot in the plat	Very High	Authoritative source. May require OCR or manual digitization for older plats.
3	County Parcel GIS Data	Parcel polygon geometry from which frontage/depth can be computed	High	Available via ArcGIS REST APIs or bulk download. Requires geometric computation.
4	Building Permit Records	Lot number and sometimes dimensions referenced in permits	Medium	Cross-reference lot numbers with plat maps to derive dimensions.
5 (Fallback)	Builder Spec Sheets / Site Plans	Lot size tiers published on builder websites or community maps	Medium	Manual collection; useful for new communities not yet in GIS.

10. Technical Considerations & Risks

10.1 Data Latency

County clerk deed recordings typically have a 2–4 week lag from closing to recording. MLS data is more current (often same-day or next-day after closing). Census building permit data has an approximate 2-month lag. The platform should clearly display the reporting period for all metrics and set user expectations for freshness.

10.2 Data Normalization

The 23 target counties maintain property data in varying formats, schemas, and update frequencies. The ETL pipeline must account for county-specific field mappings, inconsistent address formatting, and varying parcel ID conventions. A master subdivision registry should be maintained and reconciled monthly.

10.3 Subdivision Boundary Matching

Matching individual parcel sales to subdivision boundaries requires spatial joins using PostGIS. Where county GIS data includes plat boundaries, these can be used directly. Where plat boundaries are unavailable, the system should fall back to clustering parcels by legal description or subdivision name in the deed record.

10.4 Product Type Classification

Distinguishing between Detached SFH and Townhome may require a combination of property appraiser use codes, building permit descriptions, MLS property type fields, and in some cases manual classification rules per subdivision.

10.5 Lot Size Data Availability

Lot size data availability will vary significantly by subdivision and source. The platform must gracefully handle missing lot size data by displaying a clear “not available” state and still showing all other metrics. The data team should track lot size coverage rates by county and prioritize manual classification for high-activity subdivisions with missing data.

10.6 Resale Comparable Matching

The “built within 7 years” filter requires a reliable “year built” field from MLS or Zillow. In some cases, year built may be inaccurate (e.g., renovation vs. original construction). The data pipeline should cross-reference year built against Certificate of Occupancy dates where available to improve accuracy.

10.7 Builder Identification

Builder names may appear inconsistently across data sources (e.g., “DR Horton” vs. “D.R. Horton Inc.” vs. “DRHorton”). The pipeline needs a builder name normalization layer with a master builder registry that maps all known aliases to a canonical name.

10.8 Alert System Scalability

The alert/watchlist system must evaluate thresholds for every watched entity each time data refreshes. At scale, this requires an event-driven architecture (e.g., data refresh triggers an evaluation job that publishes notifications to a queue). AWS SNS + SES or a third-party service like SendGrid can handle email delivery.

11. Recommended Next Steps

15. **Data Feasibility Spike (2 weeks):** Pull sample data from 3–5 counties (e.g., Hillsborough, Orange, Duval, Brevard, Polk) and validate that subdivisions can be identified, transactions matched, lot sizes extracted, and resale comparables filtered using the proposed definitions and multi-source approach.
16. **MLS Partnership Initiation (1–2 weeks):** Engage Stellar MLS and Northeast Florida MLS to begin licensing discussions. Determine field availability for lot dimensions, concessions, and year built.
17. **Map Prototype (2 weeks):** Build a Mapbox GL JS proof-of-concept with the 23 county polygons, a working heat map toggle, click-to-drill-down for one sample county, and builder presence sidebar.
18. **Lot Size Data Audit (2 weeks):** For 5 sample subdivisions across different counties, test all 5 lot size sources (MLS, plat maps, GIS, permits, builder specs) and document coverage rates, accuracy, and ingestion complexity.
19. **API Vendor Evaluation (1 week):** Request trial access to ATTOM Data Solutions and Zillow's data products. Evaluate coverage, cost, and data granularity for the 23 target counties.
20. **Builder Name Normalization (1 week):** Compile a master builder registry for the 23-county scope, mapping all known name variants to canonical entries using DBPR contractor registry and MLS data.
21. **Product Type Classification Rules (1 week):** Define the logic for classifying properties as Detached SFH vs. Townhome using property appraiser codes, MLS property type, and permit data.
22. **Alert System Architecture (1 week):** Design the event-driven notification pipeline, define default threshold values, and select the notification delivery service.
23. **Full Architecture Design Review (1 week):** Cross-functional review with engineering, data, and product stakeholders to finalize technical architecture, sprint plan, and data source prioritization.

End of Concept Overview | March 2026